KING SALMAN INTERNATIONAL UNIVERSITY

University Requirement

All Programs

Module : Climate Change and Sustainability (ESD 217)

Lectures schedule of Part 1

- Climate Change Concept
- Climate vs. Weather
- Atmosphere layers
- Coral reef bleaching
- Climate change over years
- Climate change causes
- Climate System
- Climate indicators



Citizens in every country need to understand climate change and what it means for them and the world.

The course introduces key climate actions, such as adaptation, low carbon development, climate finance, and climate negotiations. Moreover, it introduces sustainability concept and sustainable development goals

Climate

The Climate defined as the "average weather", or as the statistical description in terms of the mean and variability of relevant quantities over a period of time ranging from months to thousands or millions of years.

Weather

The weather is the mix of events that happen each day in our atmosphere. Even though there's only one atmosphere on Earth, the weather isn't the same worldwide. Weather is different in different parts of the world and changes over minutes, hours, days, and weeks

Weather vs. Climate

- *"Weather is one football game, climate is the history of the National Football League."*
- *Climate tells you what clothes to buy, weather tells you what clothes to wear.*



Weather

Refers to short-term changes in the atmosphere



Climate

Describes what the weather is like over a long period of time in a specific area

WEATHER VS CLIMATE

SHORT-TERM STATE OF THE ATMOSPHERE

LONG-TERM PATTERN OF WEATHER



Rain





Tropical Climate



Can change within minutes or hours



Average weather over many years in one specific place

Climate vs. Weather

- *Different regions can have different climates. To describe the Climate of a place, we might say what the temperatures are like during different seasons, how windy it usually is, or how much rain or snow typically falls.*

- *When scientists talk about Climate, they're often looking at averages of precipitation, temperature, humidity, sunshine, wind, and other measures of weather that occur over a long period in a particular place.*

- *In some instances, they might look at these averages over 30 years. And, we refer to these three-decade averages of weather observations as* ***Climate Normal****.*

Climate vs. Weather

- *Weather describes what the atmosphere is doing over short time scales; it is what you see when you go outside or look out your window. Weather is extremely variable, changing from hot to cold, rainy to sunny, windy to calm from hour to hour, day to day, or season to season.*

- *Climate is the average weather pattern in a region over long periods of time, typically 30 years or more. For example, California has a Mediterranean climate; that is, historically Californians could expect long, hot, dry summers and cooler, wet winters. The weather changes with the seasons and may vary from year to year (for example, an unusually wet winter), but the general climate pattern is mostly consistent. One way to think about this is that weather is what is happening now, while climate is what you might expect to happen based on many past experiences.*

Climate vs. Weather

❖ *Climate, like weather, is variable, only on much longer time scales. Although much of the recent discussion about climate change has centered on global warming, it is important to remember that climate refers not only to temperature, but to precipitation, cloudiness, winds, and relative humidity, among other variables. As we will see, climate has fluctuated on very long-time scales in the past and will continue to do so, due to a variety of natural factors. However, current concerns over climate change relate to how human activities are causing Earth's climate to change, often referred to as anthropogenic climate change or anthropogenic global warming.*

❖ *Anthropogenic refers to the influence that humans can have on the natural world.*

❖ *Average global temperatures have been increasing since before the Industrial Revolution in the 19 century, and they will increase significantly in the future if action is not taken to reduce the emissions of greenhouse gases into the atmosphere. This is because the driving force behind this temperature increase is the increase in concentrations of greenhouse gases in the atmosphere due to human activities. more droughts while others will be subject to stronger storms.*

Examples of Anthropogenic climate change in Egypt





Egyptian Ostrich and Egyptian Ghazal


EGYPT'S RISING SEA LEVEL
THREATENS FARMLAND
WION
KSIU

Anthropogenic climate change "Worldwide"



Polar bear



Seal



Eurasian Wryneck



Climate vs. Weather

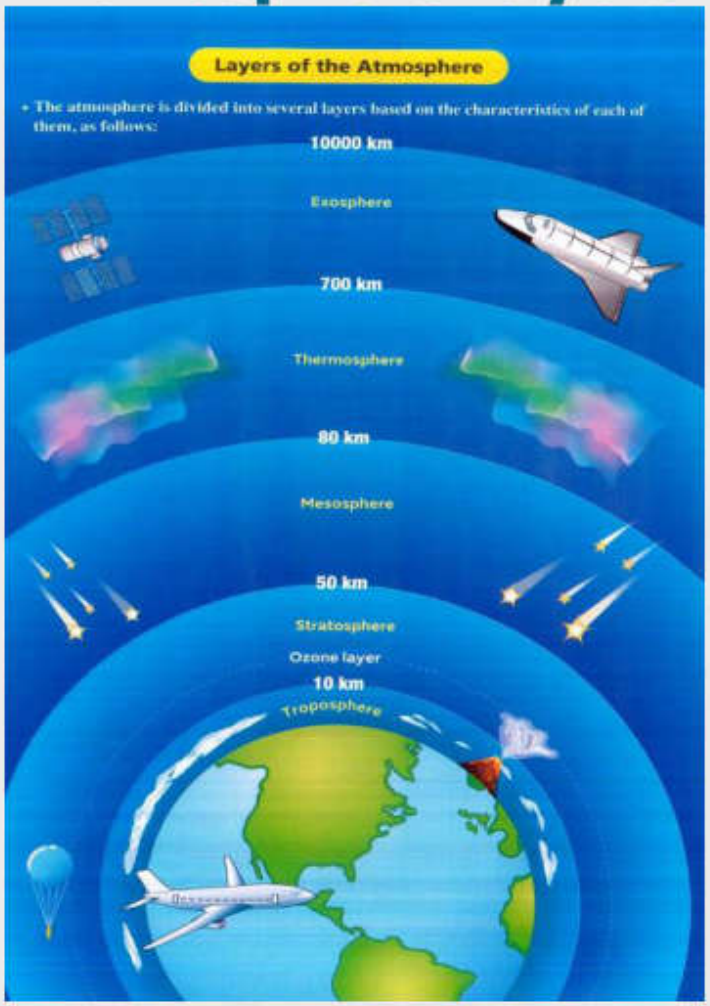
❖ *The term "climate change" is considered more accurate than "global warming," as the climate is not expected to warm uniformly around the globe: Some areas, such as the Arctic, will warm much more than others, while a few might even get cooler.*

❖*Just as important, changes in precipitation will cause some areas to experience more droughts while others will be subject to stronger storms.*

Can you distinguish Weather from Climate?

- It feels hotter than usual right now in Bali (Weather- Climate)
- Freezing temperature in Toronto for 7 days in a row (Weather- Climate)
- The dry season in Kenya is from June to October (Weather- Climate)
- Thailand has high humidity all year around (Weather- Climate)

Atmosphere Layers





Atmosphere Layers
1. Troposphere

Position

It is the closest layer to the Earth's surface in which living organisms live.
The air in this layer is denser compared to other layers.

Importance

It is the layer in which most weather and climate phenomena occur, such as cloud formation, rainfall, wind movement, and others.

Its height above Earth's surface

It extends up to approximately 10 km.

Temperature

The average temperature at Earth's surface is about 15°C.
The air temperature decreases by 6.5°C for every 1000 m (1 km) of elevation in the troposphere.

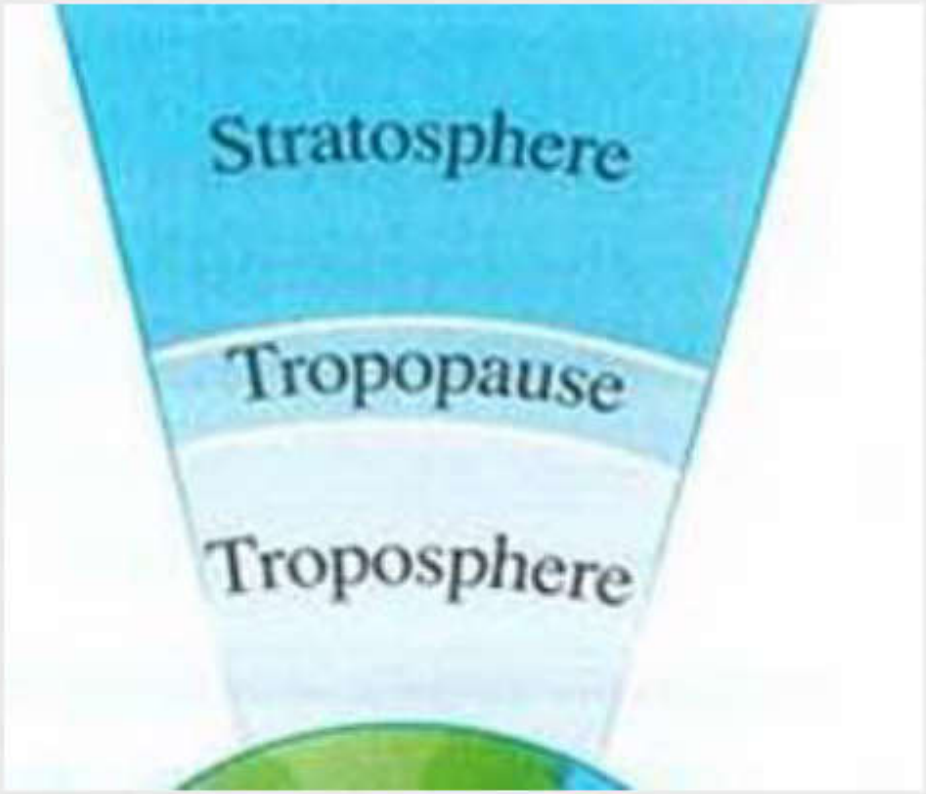
- The essential gases in the troposphere are oxygen, carbon dioxide, water vapor, and other gases provides the fundamental components necessary for life.
- The temperatures are suitable for supporting biological processes.
- The atmospheric pressure is appropriate for most living organisms.

Atmosphere Layers

Tropopause layer

- It is a thin layer of air located at the top of the troposphere, at an altitude of approximately 11 kilometers.

- It acts as a barrier or transitional zone between the troposphere and the stratosphere, where it prevents the mixing of cold air from the troposphere with the warm air in the stratosphere.



Atmosphere Layers
2. Stratosphere

Position

The layer above the troposphere (the second layer from the Earth's surface).

Its height above Earth's surface

It extends from an altitude of 10 km to about 50 km.

Temperature

It does not change through it till an altitude of 20 km from Earth's surface, then the temperature starts to rise as we go higher due to the presence of ozone layer that extends up to about 40 km as ozone gas absorbs the ultraviolet rays (UV) from solar radiation.

Importance

- Contains the ozone layer, that protects the living organisms on Earth from the harmful effects of the short-wave ultraviolet radiation.
- It is considered the preferred layer for aircraft flying since air currents movement in it is horizontal.

Atmosphere Layers
3. Mesosphere

Position

The layer above the Stratosphere (the third layer from the Earth's surface).

Its height above Earth's surface

It extends from an altitude of 50 km to 80 km.

Temperature

It is considered the coldest layer of the atmosphere, with the lowest temperatures that reach −93°C at its top. This is due to the low number of gas molecules capable of absorbing solar radiation.

Importance

Most meteors falling from space burn up and disintegrate after entering the Earth's atmosphere as they pass through this layer, which makes it play a vital role in protecting life on Earth.

Atmosphere Layers
4. Thermosphere

Position

The layer above the Mesosphere (the fourth layer from the Earth's surface).

Its height above Earth's surface

It extends from an altitude of 80 km to 700 km.

Temperature

Temperatures reach higher than 2000°C, where both oxygen and nitrogen gases absorb most of the high frequency solar radiation such as X-rays and gamma rays.

- The lower part of thermosphere, from an altitude of 80 km to 550 km called Ionosphere

Importance

It is used in long-distance radio communications due to its ability to reflect radio waves.

Atmosphere Layers
5. Exosphere

Position

The highest layer in the atmosphere

Its height above Earth's surface

It extends from an altitude of 700 km to 10000 km.

Characteristics

In this layer, some atoms and molecules move over great distances without colliding with other atoms or molecules, which gives them tremendous speeds that may allow them to overcome Earth's gravity and escape into outer space.

Importance

It is considered the ideal region for placing satellites and spacecraft, as they face almost no resistance, which allows them to remain in their orbits for long periods, and this is due to the extremely low air density in this layer.

